Remodeling LOINC in OCL

How we will use LOINC Release Files for OCL

Based on LOINC version 2.80 - may update over time

Priorities

1: Core Remodel
Main LOINC source in OCL

- Concepts:
  - LOINCs (i.e. labs)
  - LOINC Parts, Panels, etc.
  - UMLS CUIs
  - Same content and filters as search.loinc.org
  - Leverages OCL's Class, Datatype, and CodeSystem Properties(?)
  - Synonyms/translations
- Mappings:
  - Panel Structure
  - Has Answer
  - Ask Order Entry + Order Obs
  - Code Evolution
  - Associated Obs
- Hierarchy: Component by System (i.e. Grouper concepts)
- Basic release pipeline and major changes documented

2: Enrichment
Value Sets + More Mappings

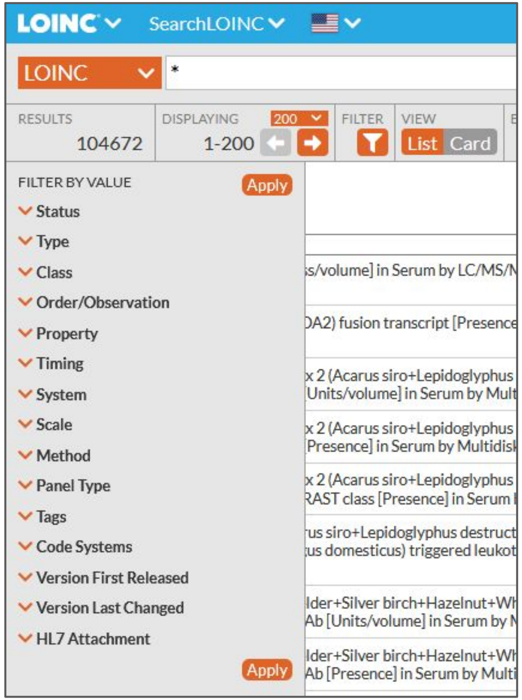
- Value Sets:
  - Answer Lists (as a value set)
  - Public Health value sets, Top 2000 + Top 20k LOINCs, etc.
  - LOINC Groups
  - Other specialized collections e.g. Doc. Ontology, Radiology
- Mappings:
  - Mappings for all 6 axes e.g. LOINC code → LOINC Parts
  - Mappings to external terminologies e.g. RadLex, RxNorm, SNOMED CT
- Optimization for FHIR
- Improved release pipeline documented

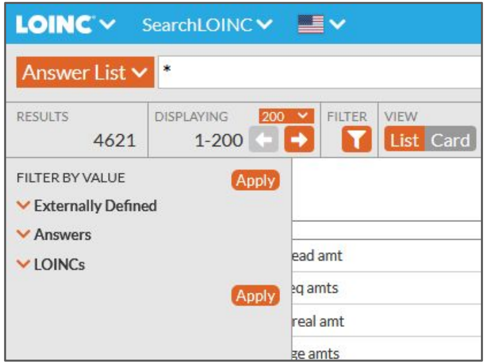
3: Other
Anything Else?

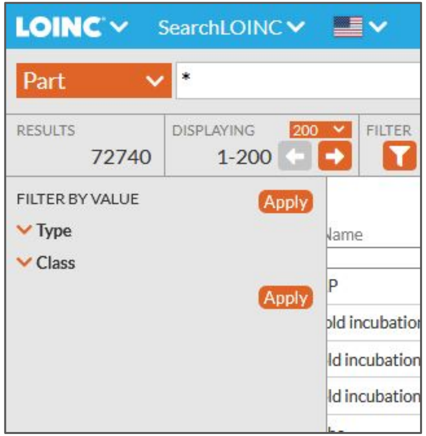
- SNOMED Collaboration Features - enhanced ontology support and additional hierarchies (?)
  -
  -
- Consumer-friendly names?
  -
- Community Mappings?
- LOINC Groups
-

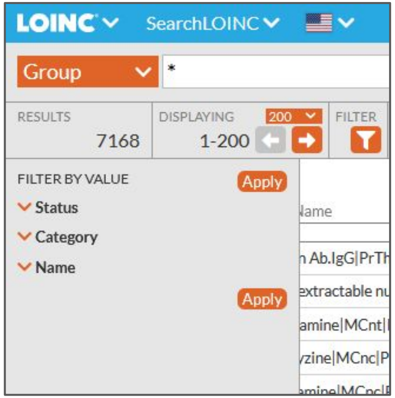
Concept Model

LOINC Code Types plus their filters









OCL does not currently have Groups but does have all the others.

Decision: LOINCs, Answer Lists + Answers, and Parts are in scope. Groups are out.

Modeling Ideas: See FHIR

LOINC FHIR: https://fhir.loinc.org/

tx.fhir.org: See postman

Using OCL's Concept Class

Directly apply LOINC Classes? https://loinc.org/kb/users-guide/classes/

What about Type?

Examples with Class Name and Code:

- Pathology [PATH]
- Histology [PATH.HISTO]
- Pathology protocols - general [PATH.PROTOCOLS.GENER]
- Urinalysis [UA]

Note: There are 4 categories of Classes (Clinical, Laboratory, Attachment, Survey) for LOINCs.

Parts have their own types. Answers, Lists, etc. are just Answers, Lists, etc.

~~Should we use the Class Name or the **Class code**? (Might be an OCL property question)? Do we need to incorporate the Class category somehow (maybe just as its own OCL property)?~~

~~Do a Type → Class Join (using codes not fully spelled out names), except for Answer List and Answers, which are their own type.~~

Decision: OCL's Concept Class will be Code Type (e.g. LOINC, Part, Answer List, etc.)

Using OCL's Datatype field

Directly apply LOINC Scale? Or map to common OCL datatypes?

Examples from LOINC with name + code plus potential OCL datatype mapping:

- Quantitative [Qn] → Numeric
- Semi-Quantitative [SemiQn] → ???
- Ordinal [Ord] → Coded
- Quantitative or Ordinal [OrdQn] → ???
- Nominal [Nom] → Coded
- Narrative [Nar] → Text
- "Multi" [Multi] → Complex
- Document [Doc] → Document
- Set [Set] → None

Unusual Cases are possible!

Birth date [21112-8] is a date but is "Qn" in LOINC.

Note: LOINC Parts, Answer Lists, Answers, and Groups likely would not have a datatype i.e. "N/A"

Decision: Use LOINC Scale as OCL's Datatype, verbatim

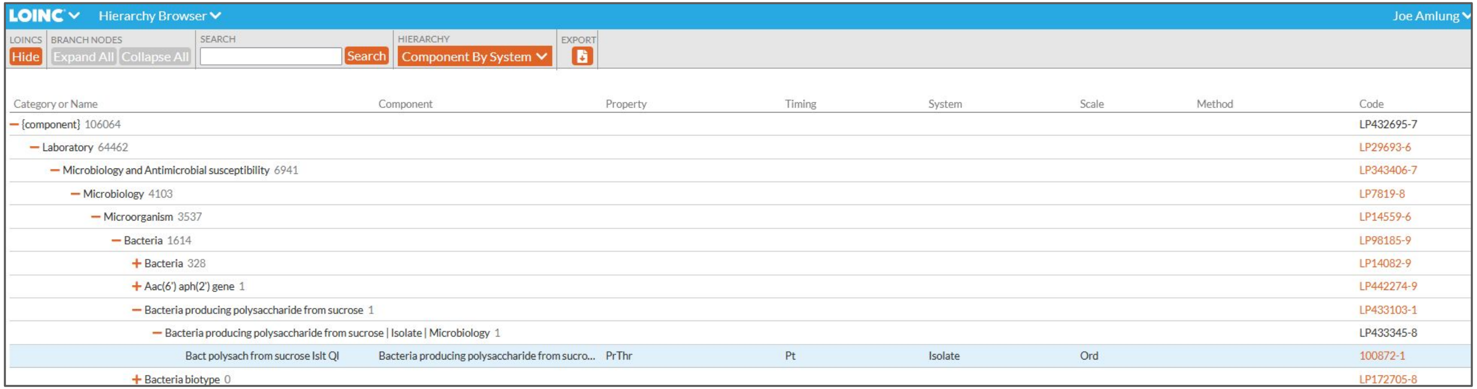
Dealing with Deprecation

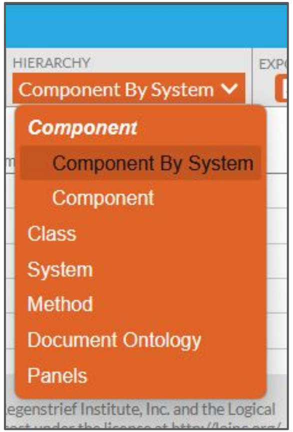
OCL has Retired (True or False)

LOINC has Active, Deprecated, Discouraged, Trial

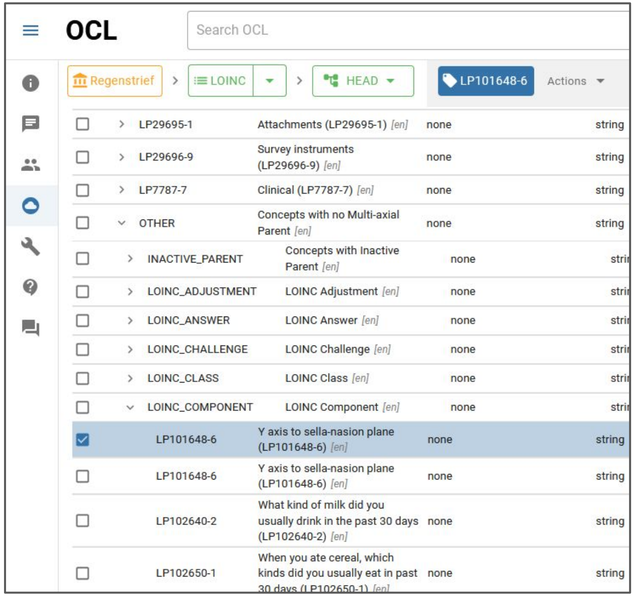
Decision: Deprecated is the only value where retired=true

Hierarchy





Note: Many pieces of LOINC don't have a multi-axial parent (e.g. Answers, some parts like [LP101648-6]) and will have to get their own "other" concept (see OCL on right)



Decision: For now, solely using Component-by-System Hierarchy that LOINC uses. This has a dedicated file in the LOINC release. Will not replicate the "Hide LOINCs" feature. Some relationships will be represented as mappings instead of hierarchy (see this slide).

Documents are not the priority, but eventually want to consider what document ontology looks like here.

Concept Names

Decision: Use Long Common Name as OCL's preferred display name

Short Name will be given as additional name.

Why not LOINC's Display Name: Display Name is not always filled, but Long Common Name is.

Consumer and Display Name: Out of scope for round 1.

LOINC CODE	LONG COMMON NAME	LOINC STATUS
102052-8	1-Deoxysphingosine [Mass/volume] in Serum by LC/MS/MS	Active

Part Description

LP435914-9 **(2S,3R)-2-Amino-4-octadecene-3-ol**
Deoxy-sphingolipids are elevated in patients with steatosis compared to those without fatty liver. Those with or without steatoses there is no difference between the different NAFLD subtypes. PMID: 33084982
Source: PubMed Central

Fully-Specified Name

Component	(2S,3R)-2-Amino-4-octadecene-3-ol
Property	MCnc
Time	Pt
System	Ser
Scale	Qn
Method	LC/MS/MS

Additional Names

Long Common Name	1-Deoxysphingosine [Mass/volume] in Serum by LC/MS/MS
Short Name	1-Deoxysphingosine Ser LC/MS/MS-mCnc
Display Name	1-Deoxysphingosine LC/MS/MS (S) [Mass/Vol]
Consumer Name ALPHA ?	1-Deoxysphingosine, Blood

Synonyms and Translations

Are all linguistic variations structured the same way?

Any languages that are weird or we otherwise need to look out for?

Decision - Names to prioritize (don’t use others e.g. Shortname):

1. FSN (i.e. the concatenation of all parts with “:” delimited, following LOINC norms)
2. Long Common Name

How we’re using OCL Properties

OCL is implementing support for Properties in Sources (see ticket #2159 and child tickets). This allows us to define source properties that can be used for searching and filtering.

Note: This model puts all types of LOINC concepts into the same source, and properties are defined at the source level. Many source properties will not apply to every individual concept, but those properties will only appear on the concept if defined as an “extra attribute”.


```
"properties": [
  {
    "code": "concept_class",
    "type": "code",
    "description": "Type of concept",
    "include_in_concept_summary": true
  },
  {
    "code": "datatype",
    "type": "code",
    "description": "Type of data captured for this concept",
    "include_in_concept_summary": true
  },
  {
    "code": "units",
    "type": "string",
    "description": "Units of measurement",
    "include_in_concept_summary": true
```

LOINC Properties in OCL

Properties (All LOINC fields as searchable attributes)

Core LOINC: status, type, class, property, timing, system, scale, method, panel_type, component, hl7_attachment, version_first_released, version_last_changed

Answer Lists: externally_defined, external_code_system, external_link, answer_list_oid, anchored_to, answers, loincs

Additional: tags, code_systems, answer_list_id, associated_observations, ask_at_order_entry, common_panel_loinc, test_loinc_num

Filters (Mirror search.loinc.org interface)

Primary Search: status, type, class, order_observation, property, timing, system, scale, method, panel_type

Answer Lists: externally_defined (Y/N), answers (count ranges), loincs (count ranges)

Additional: tags, code_systems, hl7_attachment, version_first_released, version_last_changed

Mapping Model

Mapping Option Overview

Type of Mapping	In Scope for Round 1?	Relevant LOINC Release File(s)
Panel-to-test mappings	Y	PanelsAndForms.csv
Question-to-answer mappings	Y	LoincAnswerListLink.csv (which one?)
Ask at Order Entry Mappings (i.e. Test Codes to a Common question panel)	Y	Field from main LOINC table
Code evolution (i.e. Old Concept maps to New Concept)	Y	LOINC’s main MapTo.csv
Associated Observations (i.e. Related Clinical Measures)	Y	Field from main LOINC table
Mapping to a LOINC from each axis Part (Components, Property, Time, Scale, Method, System)	N	LoincPartLink_Primary.csv LoincPartLink_Supplementary.csv?
Part-to-Part mapping (would this replicate hierarchy?)	N	LoincPartLink_Primary.csv LoincPartLink_Supplementary.csv?

Process Flow Summary

Phase	Input Files	Process	Output
1. Data Loading	All LOINC CSV files	• File validation • CSV parsing • Build lookup tables	• Loaded data structures • Lookup tables • Data quality report
2. Concept Creation	• Loinc.csv • Part.csv • AnswerList.csv • LinguisticVariants/*.csv	• Transform LOINC terms • Transform Parts • Transform Answer Lists • Transform Individual Answers • Add linguistic variants	• ~180K OCL Concept objects • Container concepts for orphans • Multi-language names
3. Mapping Creation	• Main LOINC Table • PanelsAndForms.csv • LoincAnswerListLink.csv • System field lookups • LoincPartLink_Primary.csv • LoincPartLink_Supplementary.csv?	• Panel structure mappings • Question-answer mappings • Code evolution • Order entry • Associated observations	• ~50K+ OCL Mapping objects • Various map types
4. Hierarchy Creation	ComponentHierarchyBySystem.csv	• Parse PATH_TO_ROOT • Set parent_concept_urls • Create container concepts	• Hierarchy relationships • Updated concept objects
5. UMLS Enhancement	External UMLS API	• Query API per concept • Handle rate limiting • Add CUIs to extras • Error handling	• Enhanced concepts with CUIs • API call logs
6. Output Generation	All created objects	• JSON Lines formatting • Test Set Creation • File chunking • Validation • Metadata generation	• Representative sample of concepts/mappings for test loading • Chunked .jsonl files • Import-ready format • Validation reports

Mapping out the Transformation Rules

LOINC Data Model → OCL Data Model

LOINC Terms → OCL Concepts (Source File: ‘Loinc.csv’)

LOINC Field	OCL Field	Rules
LOINC_NUM	`id`	Direct mapping - use as unique identifier
(fixed)	`concept_class`	Always set to "LOINC"
SCALE_TYP	`datatype`	Direct mapping - use LOINC Scale field
STATUS	`retired`	ACTIVE/TRIAL/DISCOURAGED → false, DEPRECATED → true
(fixed)	`source`	Always set to "LOINC"
(fixed)	`owner_type`	Always set to "Organization"
(fixed)	`owner`	Always set to "Regenstrief-Institute"
LONG_COMMON_NAME	`names[0].name`	Primary name, `name_type: "Fully Specified"`, `locale_preferred: true`
SHORTNAME	`names[1].name`	Secondary name, `name_type: "Short"`, `locale_preferred: false`
(fixed)	extras["Code Type"]	Always set to “LOINC”

Continued

LOINC Terms → OCL Concepts (Source File: ‘Loinc.csv’)

LOINC Field	OCL Field	Rules
LOINC_NUM	extras["Code in Source"]	Direct mapping
LONG_COMMON_NAME	extras["Long Common Name"]	Direct mapping
SHORTNAME	extras["Short Name"]	Direct mapping
COMPONENT	`extras["Component"]`	Direct mapping
PROPERTY	`extras["Property"]`	Direct mapping
TIME_ASPCT	`extras["Time Aspect"]`	Direct mapping
SYSTEM	`extras["System"]`	Direct mapping
SCALE_TYP	`extras["Scale"]`	Direct mapping
METHOD_TYP	`extras["Method"]`	Direct mapping
CLASS	`extras["Class"]`	Direct mapping
STATUS	`extras["Status"]`	Direct mapping

Continued

LOINC Terms → OCL Concepts (Source File: ‘Loinc.csv’)

LOINC Field	**OCL Field**	**Rules**
VersionFirstReleased	`extras["Version First Released"]`	Direct mapping
VersionLastChanged	`extras["Version Last Changed"]`	Direct mapping
ORDER_OBS	`extras["Order Obs"]`	Direct mapping
CDISC_COMMON_TESTS	`extras["CDISC Common Tests"]`	Direct mapping
HL7_FIELD_SUBFIELD_ID	`extras["HL7 Field Subfield ID"]`	Direct mapping
EXTERNAL_COPYRIGHT_NOTICE	`extras["External Copyright Notice"]`	Direct mapping
EXAMPLE_UNITS	`extras["Example Units"]`	Direct mapping

Continued

LOINC Terms → OCL Concepts (Source File: ‘Loinc.csv’)

LOINC Field	OCL Field	Rules
VersionFirstReleased	`extras["Version First Released"]`	Direct mapping
SURVEY_QUEST_TEXT	extras["Survey Quest Text"]	Direct mapping
SURVEY_QUEST_SRC	extras["Survey Quest Src"]	Direct mapping
UNITSREQUIRED	extras["Units Required"]	Direct mapping
RELATEDNAMES2	extras["Related Names 2"]	Direct mapping
DEFINITION_DESCRIPTION	extras["Definition Description"]	Direct mapping
(External API)	extras["UMLS CUI"]	Query UMLS API with LOINC_NUM, add if found

LOINC Parts → OCL Concepts (Source File: Part.csv')

LOINC Field	**OCL Field**	**Rules**
PartNumber	id	Direct mapping - use as unique identifier
(fixed)	concept_class	Always set to "LOINC Part"
(fixed)	datatype	Always set to "N/A"
Status	retired	ACTIVE → false, DEPRECATED/INACTIVE → true
(fixed)	source	Always set to "LOINC"
(fixed)	owner_type	Always set to "Organization"
(fixed)	owner	Always set to "Regenstrief-Institute"
PartDisplayName OR PartName	names[0].name	Use PartDisplayName if available, fallback to PartName
(fixed)	extras["Code Type"]	Always set to "Part"

Continued

LOINC Parts → OCL Concepts (Source File: Part.csv')

LOINC Field	OCL Field	Rules
PartNumber	extras["Code in Source"]	Direct mapping
PartName	extras["Part Name"]	Direct mapping
PartDisplayName	extras["Part Display Name"]	Direct mapping
PartTypeName	extras["Part Type"]	Direct mapping
Status	extras["Status"]	Direct mapping
ExtCodeId	extras["External Code ID"]	Direct mapping (if present)
(Hierarchy Logic)	parent_concept_urls	Set to container concepts for orphaned parts

Container Concepts Created to hold orphaned concepts:

- LOINC_COMPONENT (for Component parts without hierarchy)
- LOINC_PROPERTY (for Property parts without hierarchy)
- LOINC_TIME (for Time parts without hierarchy)
- LOINC_SYSTEM (for System parts without hierarchy)
- LOINC_SCALE (for Scale parts without hierarchy)
- LOINC_METHOD (for Method parts without hierarchy)
- OTHER (for other orphaned concepts)

Linguistic Variants → Additional OCL Concept Names (Source File: LinguisticVariants/*.csv - multiple languages)

LOINC Field	OCL Field	Rules
LOINC_NUM	(concept lookup)	Match to existing concept created in previous slide
LinguisticVariantDisplayName	names[n].name	Add as additional name entry
(Language from filename)	names[n].locale	Extract language code from file (e.g., "es", "fr", "de")
(fixed)	names[n].locale_preferred	Always set to false (English remains preferred)
(fixed)	names[n].name_type	Always set to "Fully Specified" or “Long Common Name”, based on which field it came from

Continued

LOINC Parts → OCL Concepts (Source File: Part.csv')

LOINC Field	OCL Field	Rules
PartNumber	extras["Code in Source"]	Direct mapping
PartName	extras["Part Name"]	Direct mapping
PartDisplayName	extras["Part Display Name"]	Direct mapping
PartTypeName	extras["Part Type"]	Direct mapping
Status	extras["Status"]	Direct mapping
ExtCodeId	extras["External Code ID"]	Direct mapping (if present)
(Hierarchy Logic)	parent_concept_urls	Set to container concepts for orphaned parts

Container Concepts Created to hold orphaned concepts:

- LOINC_COMPONENT (for Component parts without hierarchy)
- LOINC_PROPERTY (for Property parts without hierarchy)
- LOINC_TIME (for Time parts without hierarchy)
- LOINC_SYSTEM (for System parts without hierarchy)
- LOINC_SCALE (for Scale parts without hierarchy)
- LOINC_METHOD (for Method parts without hierarchy)
- OTHER (for other orphaned concepts)

Answer Lists → OCL Concepts (Source File: AnswerList.csv’)

LOINC Field	OCL Field	Rules
AnswerListId	id	Direct mapping - use as unique identifier (LL codes)
(fixed)	concept_class	Always set to "Answer List"
(fixed)	datatype	Always set to "N/A"
(fixed)	retired	Always set to false
(fixed)	source	Always set to "LOINC"
(fixed)	owner_type	Always set to "Organization"
(fixed)	owner	Always set to "Regenstrief-Institute"
AnswerListName	names[0].name	Primary name, name_type: "Fully Specified"
(fixed)	extras["Code Type"]	Always set to “Answer List”

Continued

Answer Lists → OCL Concepts (Source File: AnswerList.csv')

LOINC Field	OCL Field	Rules
AnswerListId	extras["Code in Source"]	Direct mapping
AnswerListName	extras["Answer List Name"]	Direct mapping
AnswerListOID	extras["Answer List OID"]	Direct mapping (if present)
ExtDefinedYN	extras["Externally Defined"]	Direct mapping
ExtDefinedAnswerListLinkId	extras["External Link ID"]	Direct mapping (if present)
ExtDefinedAnswerListCodeSystem	extras["External Code System"]	Direct mapping (if present)

Answers → OCL Concepts (Source File: AnswerList.csv’)

LOINC Field	**OCL Field**	**Rules**
AnswerStringId	id	Direct mapping - use as unique identifier (LA codes)
(fixed)	concept_class	Always set to "Answer"
(fixed)	datatype	Always set to "N/A"
(fixed)	retired	Always set to false
(fixed)	source	Always set to "LOINC"
(fixed)	owner_type	Always set to "Organization"
(fixed)	owner	Always set to "Regenstrief-Institute"
DisplayText	names[0].name	Primary name, name_type: "Fully Specified"
AnswerListId	parent_concept_urls	/orgs/Regenstrief-Institute/sources/LOINC/concepts/{AnswerListId}/
(fixed)	extras["Code Type"]	Always set to “Answer”

Continued

Answers → OCL Concepts (Source File: AnswerList.csv')

LOINC Field	OCL Field	Rules
AnswerStringId	extras["Code in Source"]	Direct mapping
DisplayText	extras["Display Text"]	Direct mapping
AnswerListId	extras["Answer List ID"]	Direct mapping
LocalAnswerCode	extras["Local Answer Code"]	Direct mapping
SequenceNumber	extras["Sequence"]	Direct mapping
Score	extras["Score"]	Direct mapping (if present)
ExtCodeId	extras["External Code ID"]	Direct mapping (if present)
ExtCodeDisplayName	extras["External Code Display Name"]	Direct mapping (if present)
ExtCodeSystem	extras["External Code System"]	Direct mapping (if present)

Panel Structure → OCL Mappings (Source File: PanelsAndForms.csv’)

LOINC Field	OCL Field	Rules
(fixed)	type	Always set to "Mapping"
(fixed)	map_type	Always set to "has element"
ParentLoinc	from_concept_url	/orgs/Regenstrief-Institute/sources/LOINC/concepts/{ParentLoinc}/
LoincNumber	to_concept_url	/orgs/Regenstrief-Institute/sources/LOINC/concepts/{LoincNumber}/
(fixed)	source	Always set to "LOINC"
(fixed)	owner_type	Always set to "Organization"
(fixed)	owner	Always set to "Regenstrief-Institute"
SequenceInPanel	extras["Sequence"]	Direct mapping
Required	extras["Required"]	Direct mapping
CardinalityMin + CardinalityMax	extras["Cardinality"]	Combine as "Min..Max" format
AnswerListIdOverride	extras["Answer List Override"]	Direct mapping (if present)
AnswerListTypeOverride	extras["Answer List Type Override"]	Direct mapping (if present)

Question-Answer Links → OCL Mappings (Source Files: LoincAnswerListLink.csv + AnswerList.csv using a JOIN)

LOINC Field	OCL Field	Rules
(fixed)	type	Always set to "Mapping"
(fixed)	map_type	Always set to "has answer"
LoincAnswerListLink.LoincNumber	from_concept_url	/orgs/Regenstrief-Institute/sources/LOINC/concepts/{LoincNumber}/
AnswerList.AnswerStringId	to_concept_url	/orgs/Regenstrief-Institute/sources/LOINC/concepts/{AnswerStringId}/ (Individual Answer)
(fixed)	source	Always set to "LOINC"
(fixed)	owner_type	Always set to "Organization"
(fixed)	owner	Always set to "Regenstrief-Institute"
LoincAnswerListLink.AnswerListId	extras["Answer List ID"]	Direct mapping
LoincAnswerListLink.AnswerListLinkType	extras["Answer List Type"]	Direct mapping
AnswerList.SequenceNumber	extras["Sequence"]	From JOIN on AnswerListId
AnswerList.Score	extras["Score"]	From JOIN on AnswerListId
AnswerList.LocalAnswerCode	extras["Local Answer Code"]	From JOIN on AnswerListId

JOIN Condition: `LoincAnswerListLink.AnswerListId = AnswerList.AnswerListId`

Code Evolution → OCL Mappings (Source File: MapTo.csv)

LOINC Field	OCL Field	Rules
(fixed)	type	Always set to "Mapping"
(fixed)	map_type	Always set to "Map To"
OLD_LOINC_NUM	from_concept_url	/orgs/Regenstrief-Institute/sources/LOINC/concepts/{OLD_LOINC_NUM}/
NEW_LOINC_NUM	to_concept_url	/orgs/Regenstrief-Institute/sources/LOINC/concepts/{NEW_LOINC_NUM}/
(fixed)	source	Always set to "LOINC"
(fixed)	owner_type	Always set to "Organization"
(fixed)	owner	Always set to "Regenstrief-Institute"
ChangeReason	extras["Change Reason"]	Direct mapping (if available)

Order Entry → OCL Mappings (Source File: Main LOINC table)

LOINC Field	OCL Field	Rules
(fixed)	*type*	*Always set to "Mapping"*
(fixed)	*map_type*	*Always set to "Ask At Order Entry"*
TEST_LOINC_NUM	*from_concept_url*	*/orgs/Regenstrief-Institute/sources/LOINC/concepts/{TEST_LOINC_NUM}/*
COMMON_PANEL_LOINC	*to_concept_url*	*/orgs/Regenstrief-Institute/sources/LOINC/concepts/{COMMON_PANEL_LOINC}/*
(fixed)	*source*	*Always set to "LOINC"*
(fixed)	*owner_type*	*Always set to "Organization"*
(fixed)	*owner*	*Always set to "Regenstrief-Institute"*

Associated Observations → OCL Mappings (Source File: Main LOINC table)

LOINC Field	**OCL Field**	**Rules**
(fixed)	*type*	*Always set to "Mapping"*
(fixed)	*map_type*	*Always set to "Associated Observations"*
PRIMARY_LOINC_NUM	*from_concept_url*	*/orgs/Regenstrief-Institute/sources/LOINC/concepts/{PRIMARY_LOINC_NUM}/*
RELATED_LOINC_NUM	*to_concept_url*	*/orgs/Regenstrief-Institute/sources/LOINC/concepts/{RELATED_LOINC_NUM}/*
(fixed)	*source*	*Always set to "LOINC"*
(fixed)	*owner_type*	*Always set to "Organization"*
(fixed)	*owner*	*Always set to "Regenstrief-Institute"*

Hierarchy Structure → OCL parent_concept_urls (Source File: ComponentHierarchyBySystem.csv)

LOINC Field	OCL Field	Rules
PATH_TO_ROOT	*(parse for relationships)*	*Parse dot-delimited path to identify parent-child structure*
IMMEDIATE_PARENT	*parent_concept_urls*	*/orgs/Regenstrief-Institute/sources/LOINC/concepts/{IMMEDIATE_PARENT}/*
CODE	*(target concept)*	*The concept receiving the parent relationship*
SEQUENCE	*(implicit ordering)*	*Ordering within parent (not directly mapped to OCL)*

Transformation Logic:

- If CODE is LOINC term: Set parent_concept_urls to IMMEDIATE_PARENT Part
- If CODE is LOINC Part: May have Part parent or be root-level
- Create container concepts for orphaned items without multi-axial parents

External UMLS API Integration (Source: External UMLS API)

API Input	OCL Target	Transformation Rules
LOINC_NUM (from LOINC Terms)	*extras["UMLS CUI"]*	*Query UMLS API, add CUI if found: "C1234567"*
PartNumber (from Part concepts)	*extras["UMLS CUI"]*	*Query UMLS API, add CUI if found: "C1234567"*
AnswerListId (from Answer List concepts)	*extras["UMLS CUI"]*	*Query UMLS API, add CUI if found: "C1234567"*
AnswerStringId (from Individual Answer concepts)	*extras["UMLS CUI"]*	*Query UMLS API, add CUI if found: "C1234567"*

Process Rules:

- Execute AFTER initial concept creation from LOINC files
- Handle API failures gracefully (log error, continue without CUI)
- Implement rate limiting and batch processing for efficiency
- Only add UMLS CUI to extras if successfully retrieved

Out of Scope for Round 1

❌ DEFERRED TO PHASE 2

LOINC File	Content	Reason for Deferral
`LoincPartLink_Primary.csv`	Multi-axial Part mappings	Phase 2: Enrichment
`LoincPartLink_Supplementary.csv`	Extended Part mappings	Phase 2: Enrichment
`PartRelatedCodeMapping.csv`	External terminology mappings	Phase 2: Enrichment
Consumer name files	Consumer-friendly names	Phase 3: Other
Group-related files	LOINC Groups	Phase 3: Other

Other Files Ignored

Administrative/Change Management Files

- `Updates.csv` - Change tracking (only using for Code Evolution mappings if possible)
- `UpdatesAdditionalFields.csv` - Extended change tracking (administrative detail not needed)
- `ChangeSnapshot/*.csv` - Historical changes (only using if source for Code Evolution mappings)

Specialized Domain Files

- Files in `RadiologyPlaybook/` - Radiology-specific ontology (deferred to Phase 2: Enrichment)
- Files in `DocumentOntologyOWL/` - Document ontology OWL (deferred to Phase 2: Enrichment)
- Consumer-friendly name files - Consumer naming (deferred to Phase 2: Enrichment)

Fields ignored: Loinc.csv

Technical/Administrative Fields

- `SUBMITTED_UNITS` - Technical detail not needed in OCL (used during LOINC submission process)
- `CHNG_TYPE` - Administrative tracking (internal LOINC change management)
- `CURATED_RANGE_AND_UNITS` - Complex technical field (specialized for lab systems, not core terminology)
- `DOCUMENT_SECTION` - Document-specific organization (only relevant for document ontology - Phase 2)
- `FORMULA` - Technical calculation (mathematical formulas not needed in core concepts)
- `SPECIES` - Specialized veterinary field (not in scope for initial human-focused implementation)
- `EXMPL_ANSWERS` - Example data only (not standardized terminology)
- `ACSSYM` - Legacy formatting symbol (deprecated formatting information)
- `BASE_NAME` - Internal LOINC processing (used for LOINC creation, not end-user terminology)
- `FINAL` - Internal status flag (administrative, covered by STATUS field)

Fields ignored: Loinc.csv (Continued)

Display/Formatting Fields

- `NAACCR_ID` - Specialized registry identifier (cancer registry specific, niche use case)
- `CODE_TABLE` - Internal LOINC reference (administrative reference, not terminology content)
- `SETROOT` - Internal LOINC grouping (used for LOINC maintenance, not clinical semantics)
- `PANELTYPE` - Administrative classification (panel organization handled via mappings instead)
- `ADJUSTMENTS_APPLIED` - Technical processing flag (internal LOINC creation process)

Legacy/Deprecated Fields

- `OLD_CLASS` - Historical data (superseded by current CLASS field)
- `COMMENTS` - Free text annotations (unstructured, not suitable for terminology)
- `STATUS_REASON` - Administrative text (free text, not standardized)
- `STATUS_TEXT` - Administrative text (redundant with STATUS field)
- `CHANGE_REASON_PUBLIC` - Historical tracking (administrative, not clinical content)

Fields ignored: Part.csv

Administrative Fields

- `CreatedDate` - Administrative timestamp (not relevant for terminology content)
- `ModifiedDate` - Administrative timestamp (not relevant for terminology content)
- `PartComment` - Free text annotation (unstructured comments not suitable for OCL)
- `PartSuper` - Internal LOINC hierarchy (complex internal relationships, using ComponentHierarchy instead)

Technical/Internal Fields

- `PartSequenceOrder` - Internal ordering (administrative sequencing not needed in OCL)
- `CodeSystemOID` - Technical identifier (OID not used in OCL model)
- `CodeSystemVersion` - Version tracking (handled at source level in OCL)

Fields ignored: AnswerList.csv

External System Fields

- `ExtDefinedAnswerListCodeSystemOID` - External system OID (technical identifiers not needed in core model)
- `ExtDefinedAnswerListCodeSystemVersion` - External versioning (handled differently in OCL)
- `ExtDefinedAnswerListCodeSystemName` - External system name (only keeping essential external reference)

Administrative Fields

- `AnswerListComment` - Free text comments (unstructured, not suitable for terminology)
- `CreatedDate` - Administrative timestamp (not relevant for terminology content)
- `ModifiedDate` - Administrative timestamp (not relevant for terminology content)

Fields ignored: PanelsAndForms.csv

Technical Constraints

- `PanelId` - Internal panel identifier (using LOINC codes instead for consistency)
- `Id` - Row identifier (administrative, not semantic)
- `PartName` - Redundant reference (part information retrieved from Part.csv)
- `PartNumber` - Redundant reference (part information retrieved from Part.csv)

Complex Display Logic

- `ConditionalRequired` - Complex business logic (advanced form logic not in scope for Phase 1)
- `Threshold` - Clinical decision logic (complex clinical rules deferred to Phase 2)
- `Units` - Technical specifications (unit handling not in core terminology scope)

Fields ignored: ComponentHierarchyBySystem.csv

Display/Presentation Fields

- `CODE_TEXT` - Display reference only (text retrieved from source concepts instead)
- `IsSearchable` - UI behavior flag (user interface concern, not terminology)
- `LOINC_CLASS_TYPE` - Display categorization (using direct concept classes instead)

Fields ignored: LoincAnswerListLink.csv

Administrative Fields

- `ApplicableContext` - Complex contextual rules (advanced business logic deferred to Phase 2)
- `CreatedDate` - Administrative timestamp (not relevant for terminology content)
- `ModifiedDate` - Administrative timestamp (not relevant for terminology content)

Fields ignored: LinguisticVariants/*.csv

Technical Fields

- `TranslationStatus` - Administrative status (translation workflow, not terminology content)
- `TranslatorID` - Administrative tracking (not relevant for end users)
- `ReviewStatus` - Administrative workflow (translation quality workflow)
- `LastModified` - Administrative timestamp (not relevant for terminology content)

Technical Formatting

- `DisplayFormat` - Presentation logic (user interface concern, not core terminology)
- `SortOrder` - Display ordering (UI-specific, not semantic)